Model Evaluation

Classification quality metrics, prediction error, cross-validation

Data Analytics and Mining, 2026

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Classification quality metrics

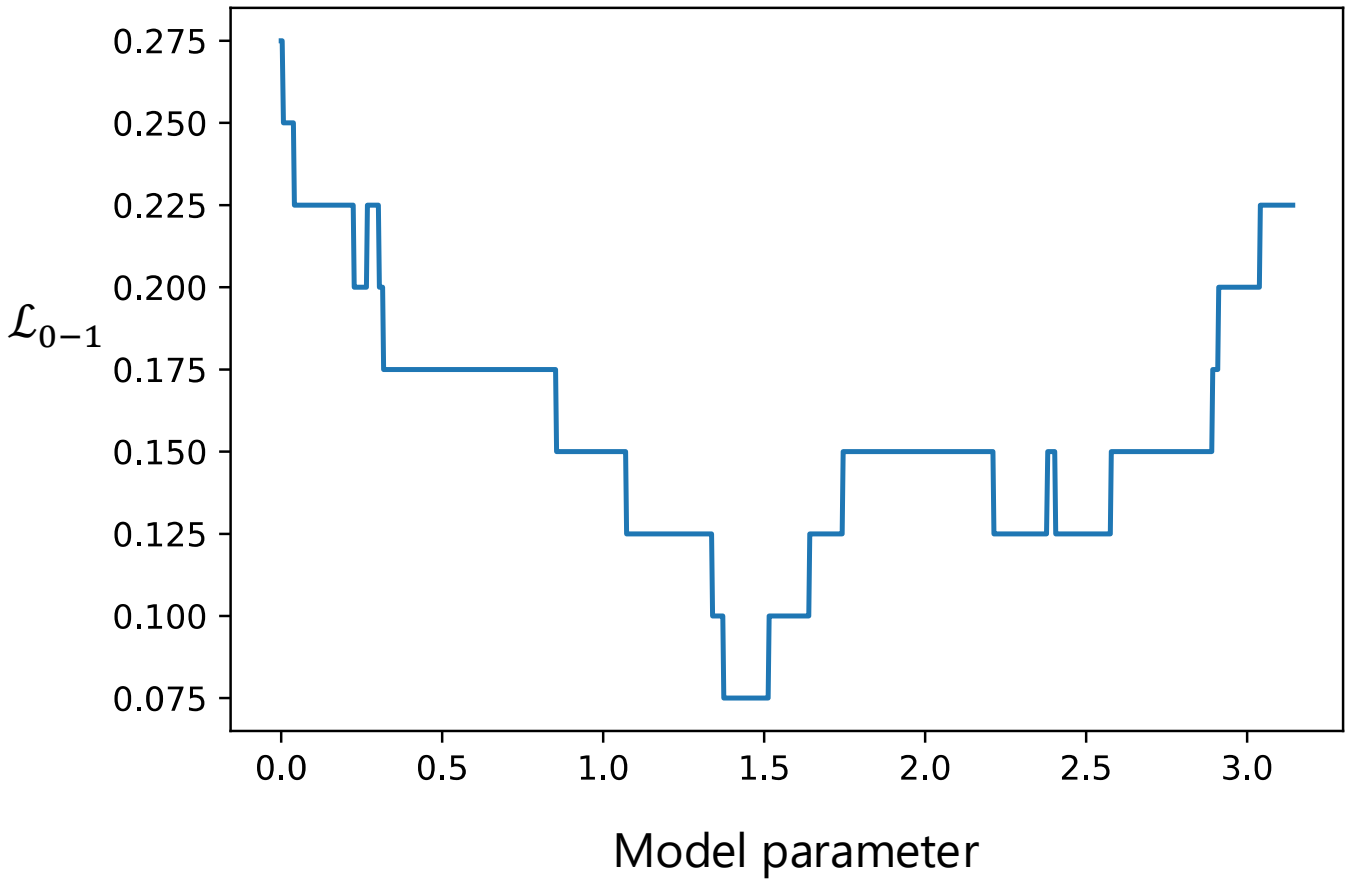


How to evaluate a classifier?

0-1 Loss

- Probability of an error (error rate):

$$\mathcal{L}_{0-1} = \frac{1}{N} \sum_{i=1\ldots N} \mathbb{I}(y_i \neq \hat{y}_i)$$



0-1 Loss

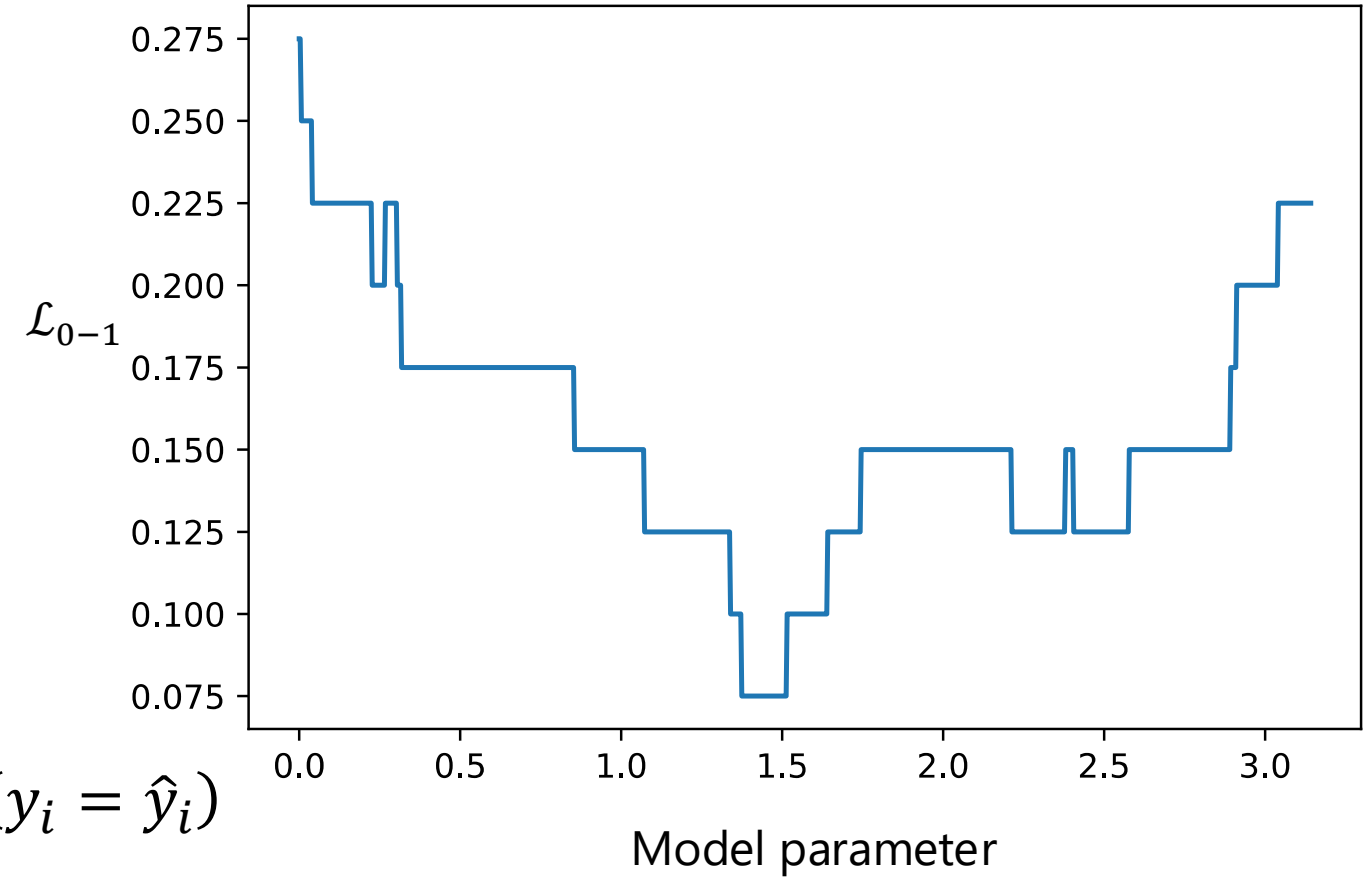
- Probability of an error (error rate):

$$\mathcal{L}_{0-1} = \frac{1}{N} \sum_{i=1\ldots N} \mathbb{I}(y_i \neq \hat{y}_i)$$

- Accuracy:

$$accuracy = 1 - \mathcal{L}_{0-1} = \frac{1}{N} \sum_{i=1\ldots N} \mathbb{I}(y_i = \hat{y}_i)$$

- Not always a good quality measure
  - E.g. when classes are imbalanced



Confusion matrix

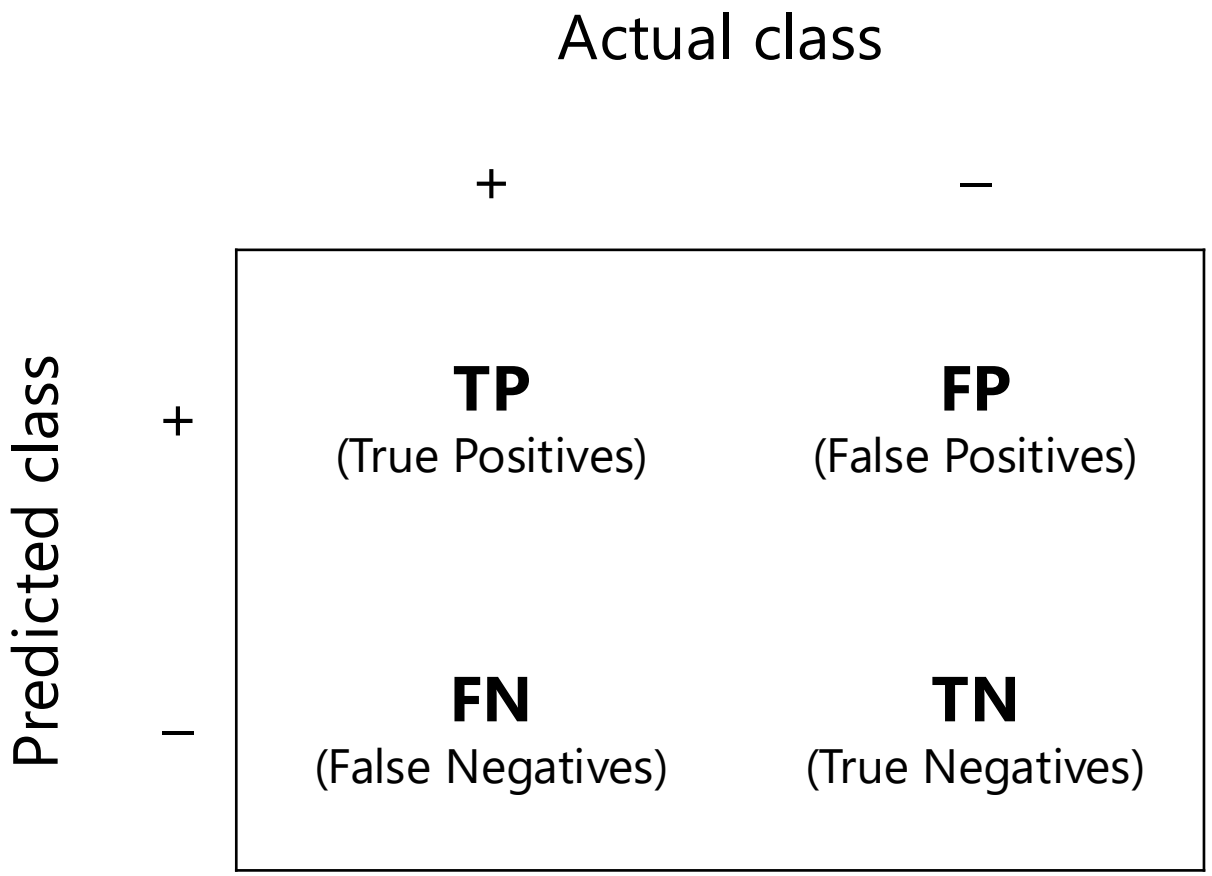
Predicted class \ Actual class	1	2	⋯	C
1	n_{11}	n_{12}	⋯	n_{1C}
2	n_{21}	n_{22}	⋯	n_{2C}
⋮	⋮	⋮	⋱	⋮
C	n_{C1}	n_{C2}	⋯	n_{CC}

n_{ij} – number of objects of class j, that were predicted as class i

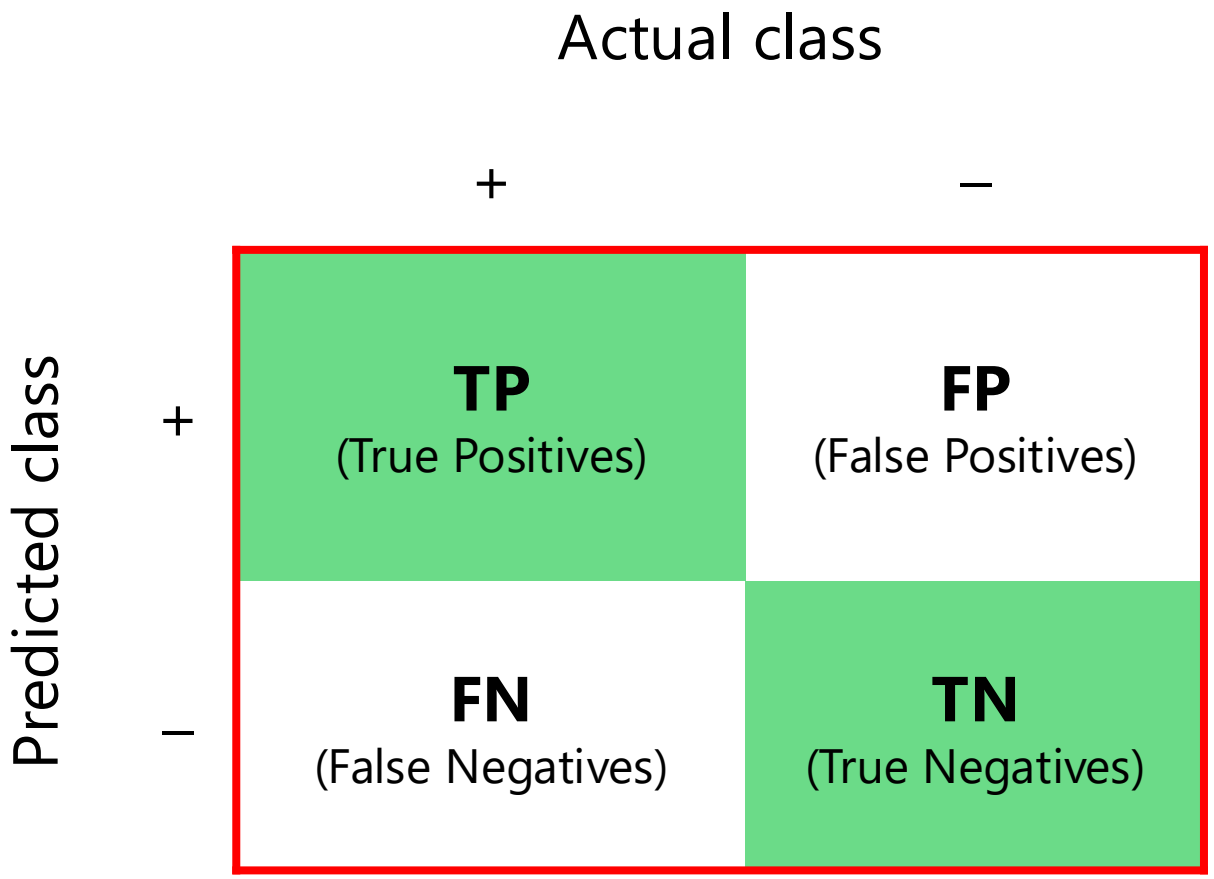
Diagonal elements – correct classifications

Off-diagonal elements – incorrect classifications

Binary case

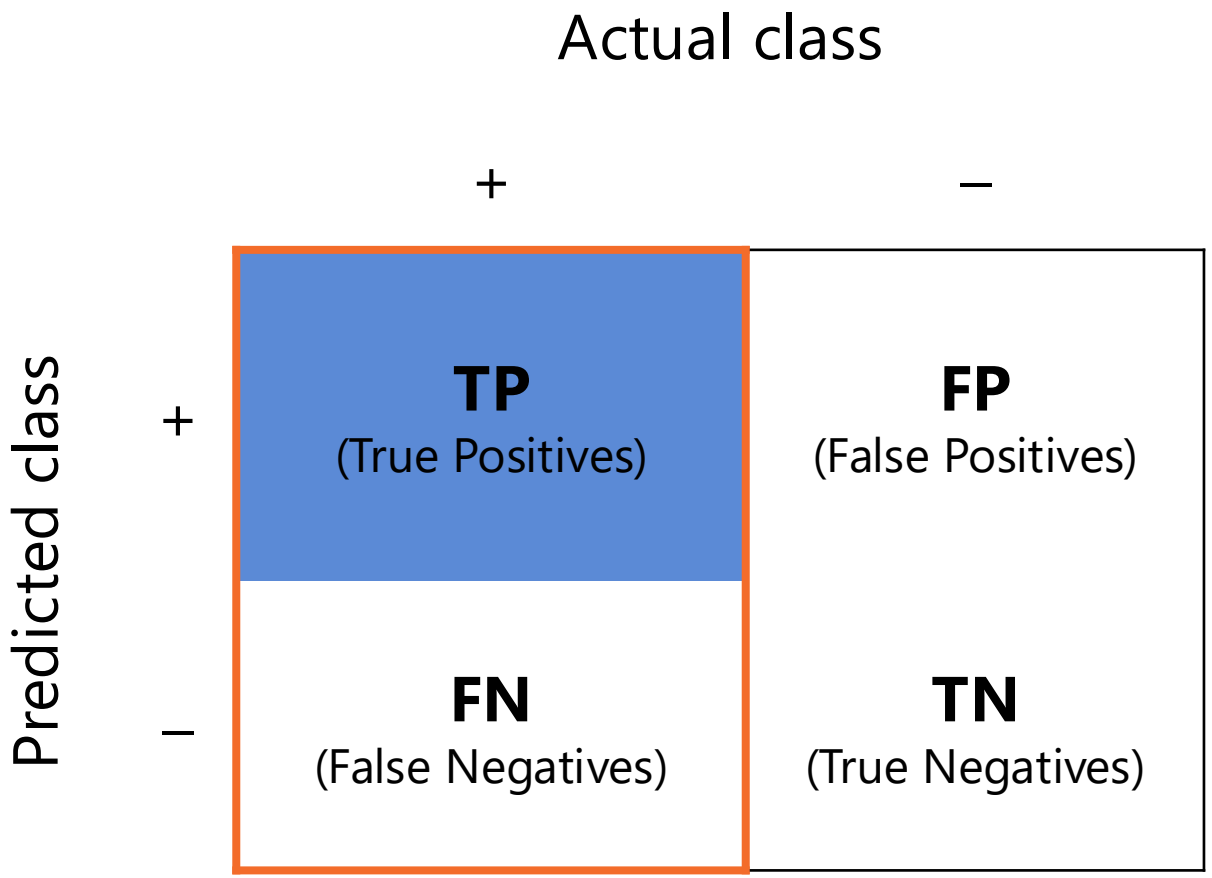

Actual class
+
–
Predicted class
+
–
TP
(True Positives)
FP
(False Positives)
FN
(False Negatives)
TN
(True Negatives)

Binary case



$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

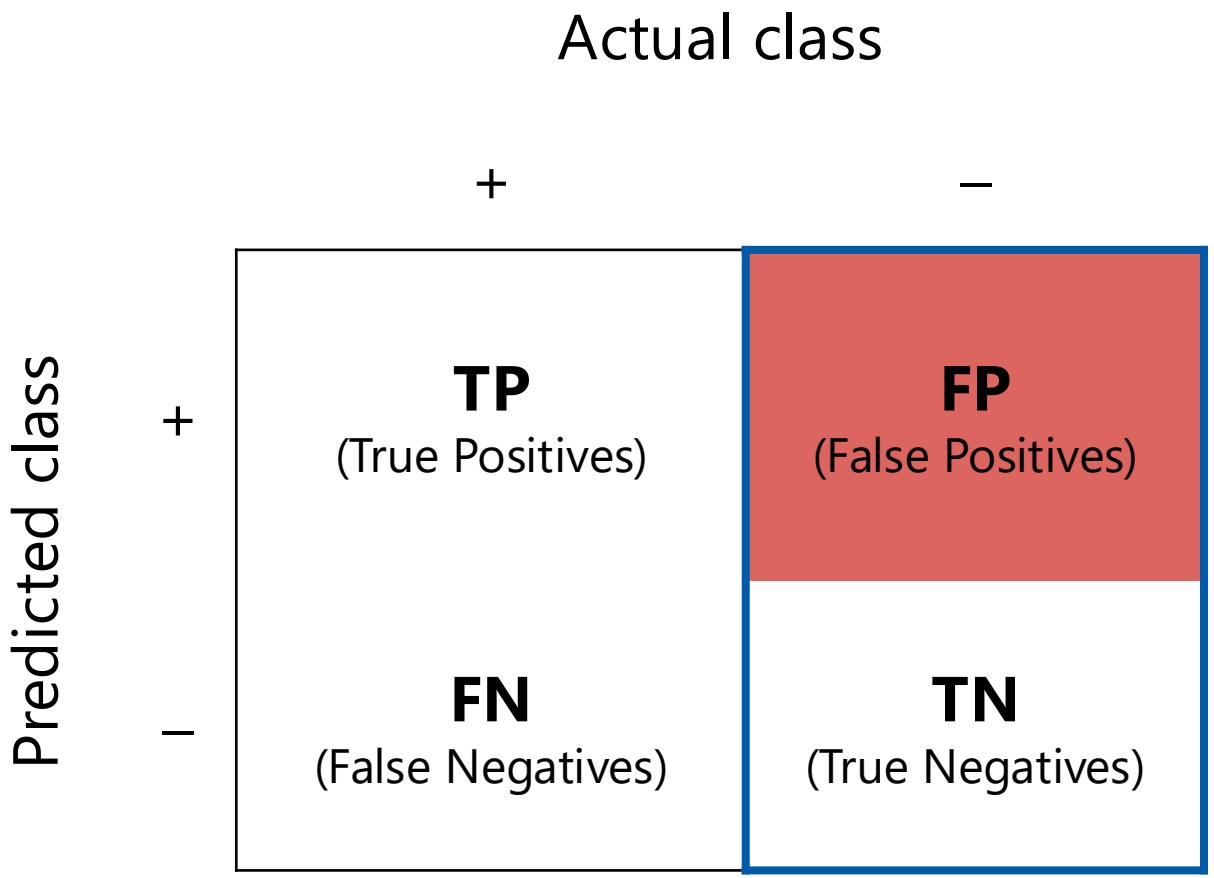
Binary case



$$\text{Accuracy} = \frac{\text{TP+TN}}{\text{TP+TN+FP+FN}}$$

$$\text{True positive rate, TPR} = \frac{\text{TP}}{\text{TP+FN}}$$

Binary case

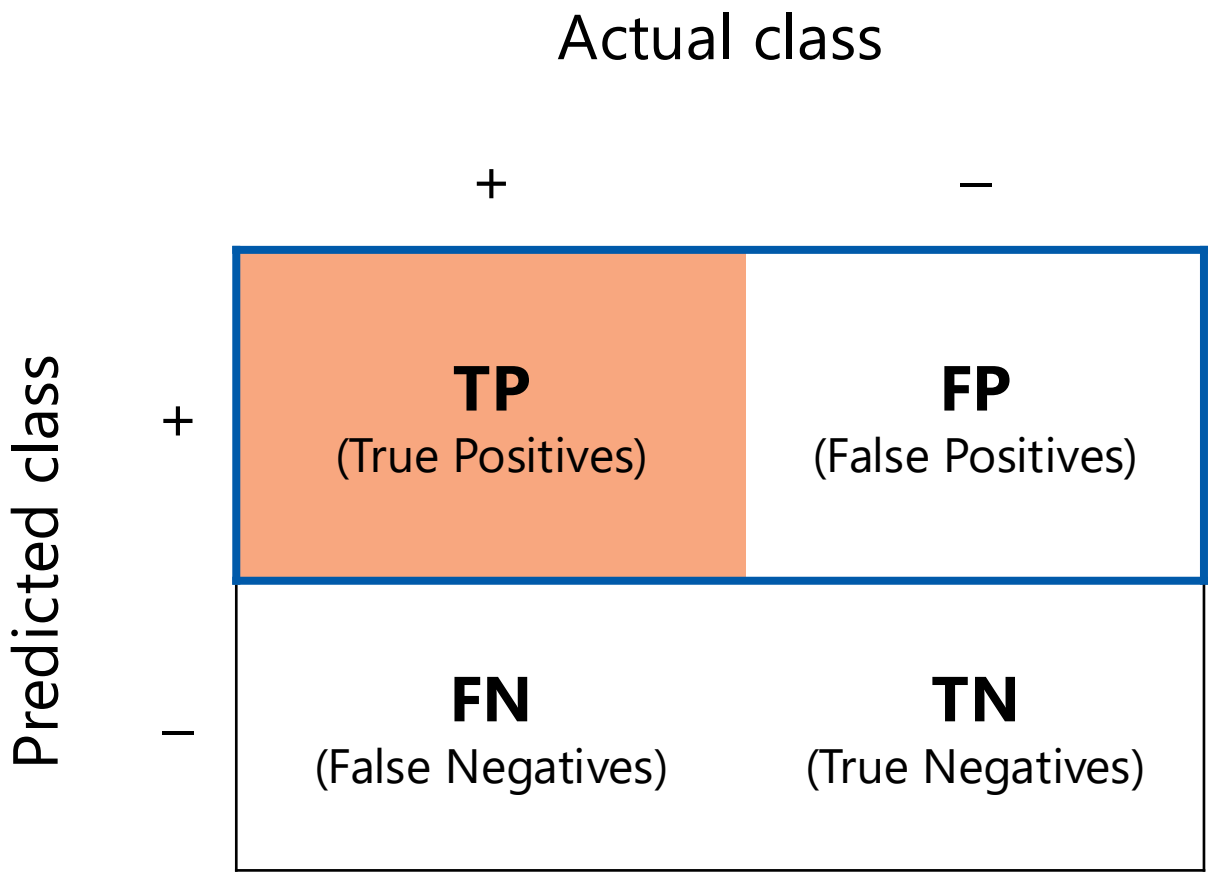


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Binary case



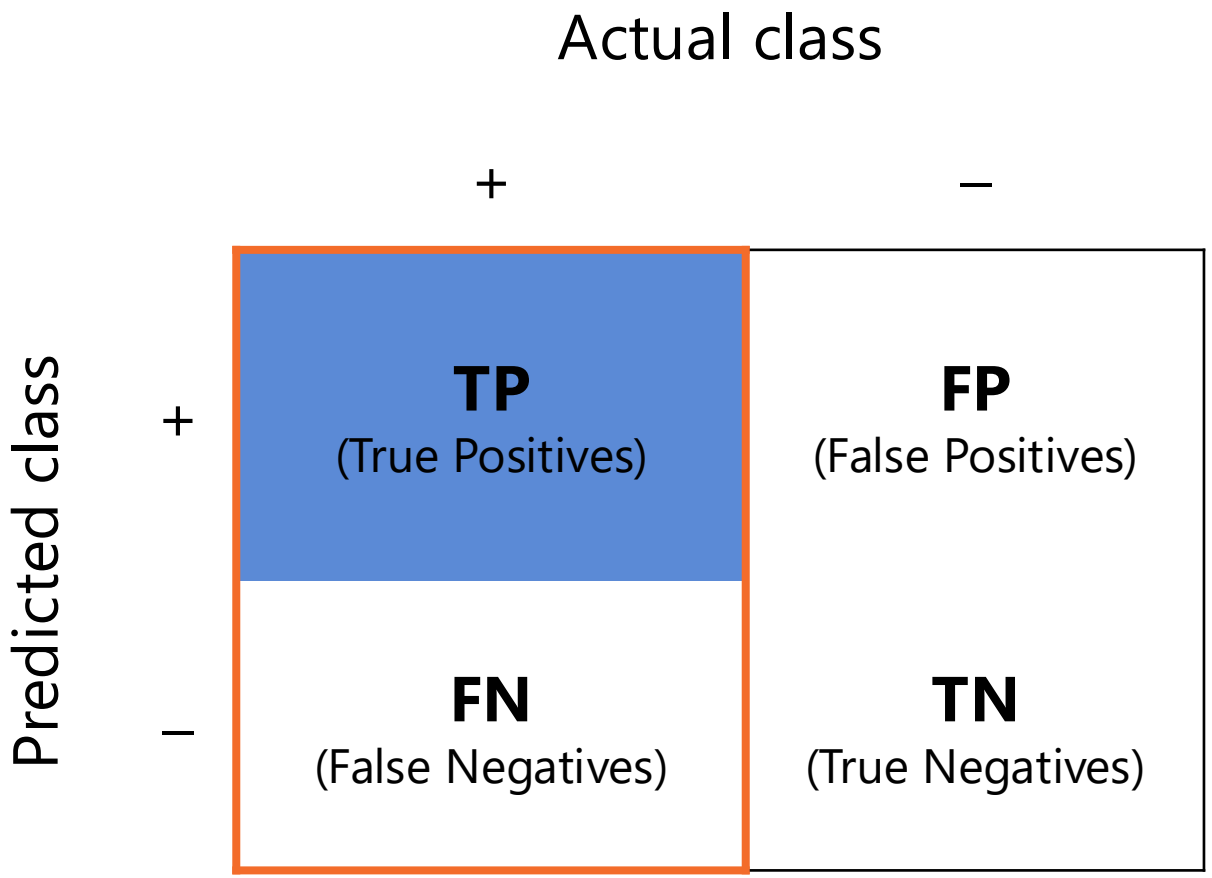
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$$\text{True positive rate, TPR} = \frac{TP}{TP+FN}$$

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$$\text{Precision} = \frac{TP}{TP+FP}$$

Binary case



$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

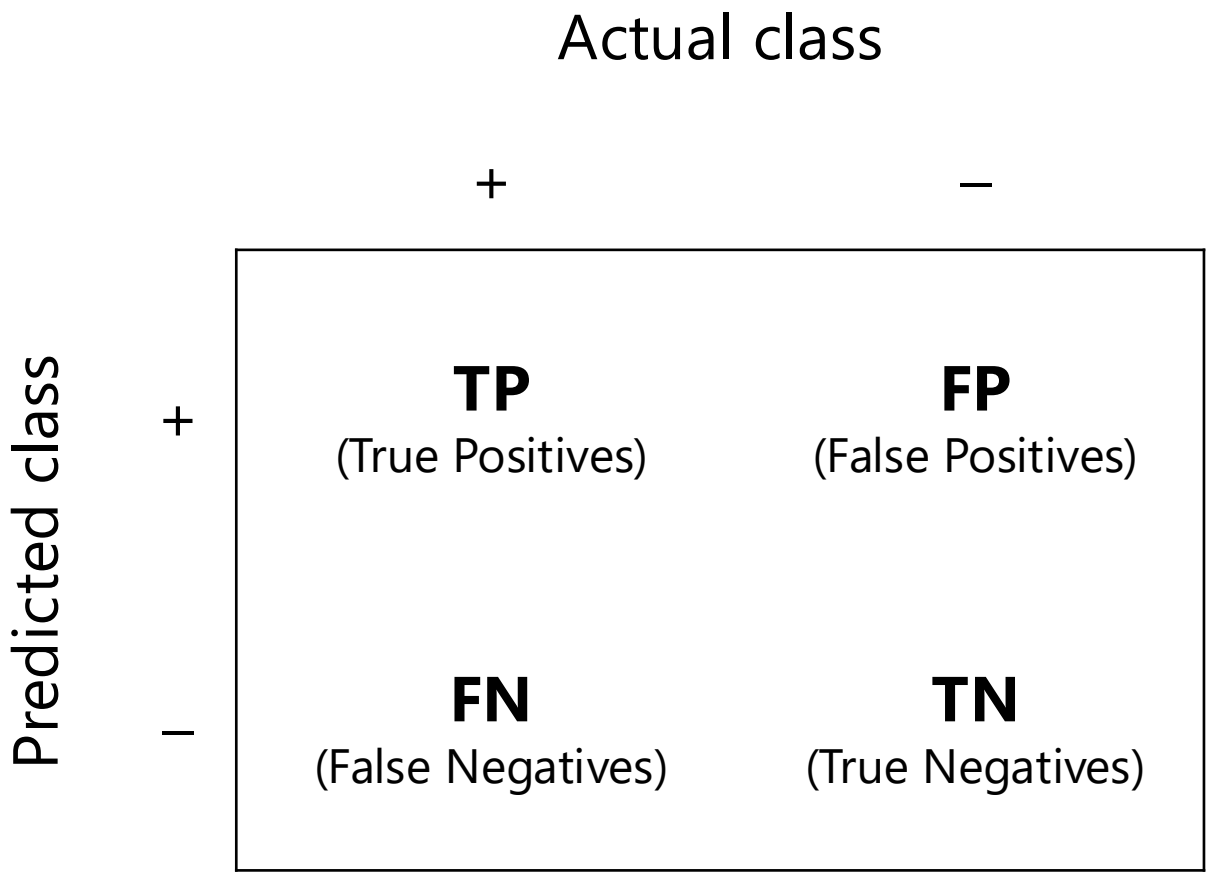
$$\text{True positive rate, TPR} = \frac{TP}{TP+FN}$$

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$$\text{Precision} = \frac{TP}{TP+FP}$$

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Binary case



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$$\text{False positive rate, FPR} = \frac{FP}{FP+TN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$F_1\text{-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Learning check

1. A classifier on 100 objects: TP = 40, FP = 10, FN = 20, TN = 30. Its accuracy is …

a. 0.4
b. 0.7
c. 0.8
d. 0.3

2. Same classifier (TP = 40, FP = 10, FN = 20, TN = 30). Its precision is …

a. 0.67
b. 0.4
c. 0.5
d. 0.8

3. Same classifier. Its recall (TPR) is …

a. 0.67
b. 0.8
c. 0.4
d. 0.75

4. Same classifier. Its false positive rate (FPR) is …

a. 0.1
b. 0.2
c. 0.25
d. 0.33

Prediction error
vs
expected prediction error



Two scenarios

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$$\mathrm{Err}_\tau = \mathop{\mathbb{E}}_{x,y}\left[L\left(y, \widehat{f_\tau}(x)\right)\right]$$

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  - **Expected prediction error**:

$$\mathrm{Err} = \mathop{\mathbb{E}}_{x,y,\tau}\left[L\left(y, \widehat{f}_\tau(x)\right)\right] = \mathop{\mathbb{E}}_{\tau}[\mathrm{Err}_\tau]$$

Splitting to train and test



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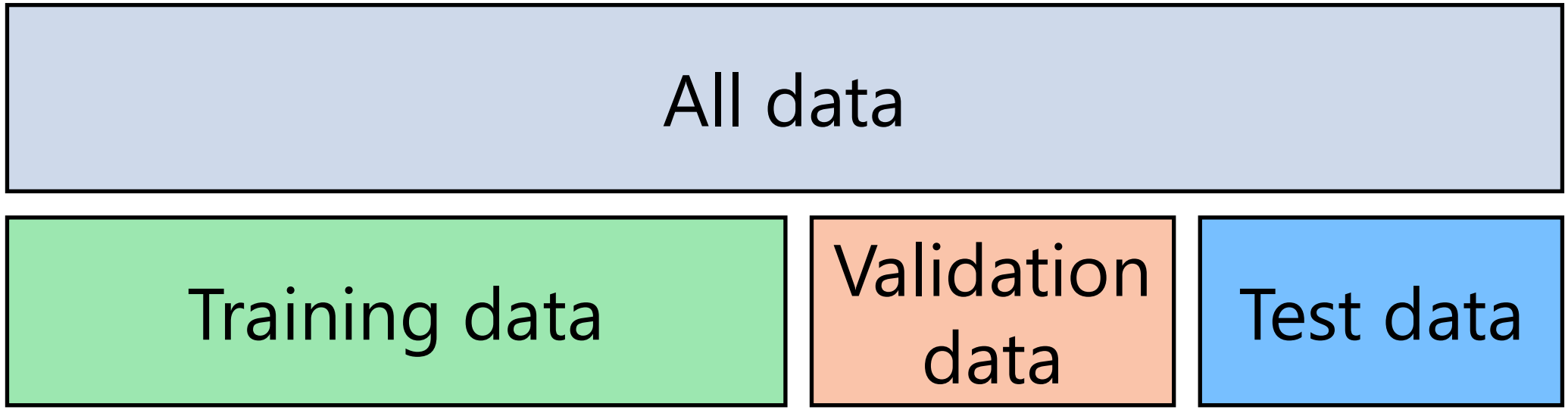
How to estimate its variance?

Splitting to train, validation and test

- When we do model selection, we use the left-out data to estimate the prediction error and minimize it (e.g., wrt the hyperparameters)
- May ‘overfit to test’, so the resulting minimized error is not a good estimate of the prediction error

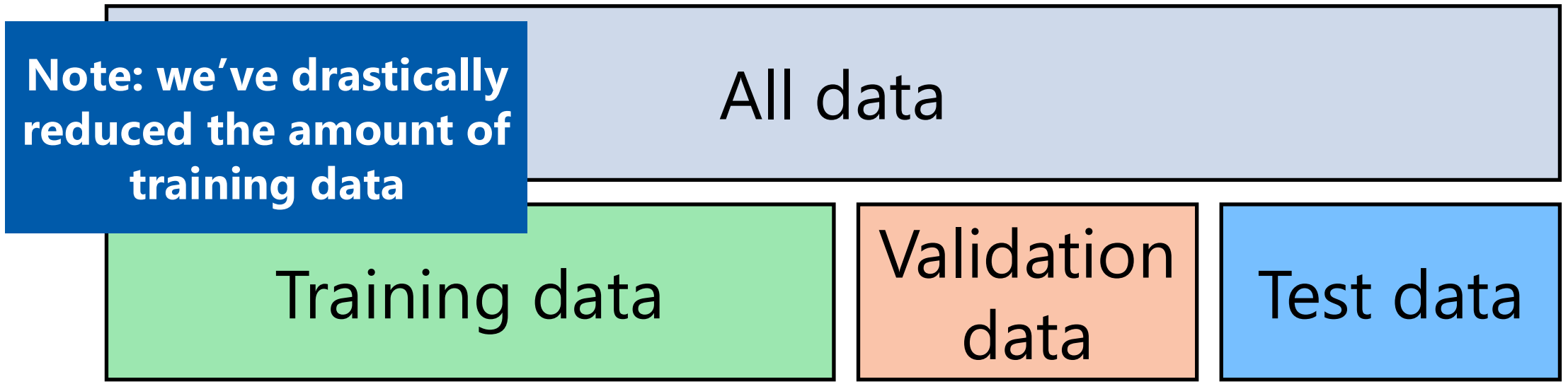
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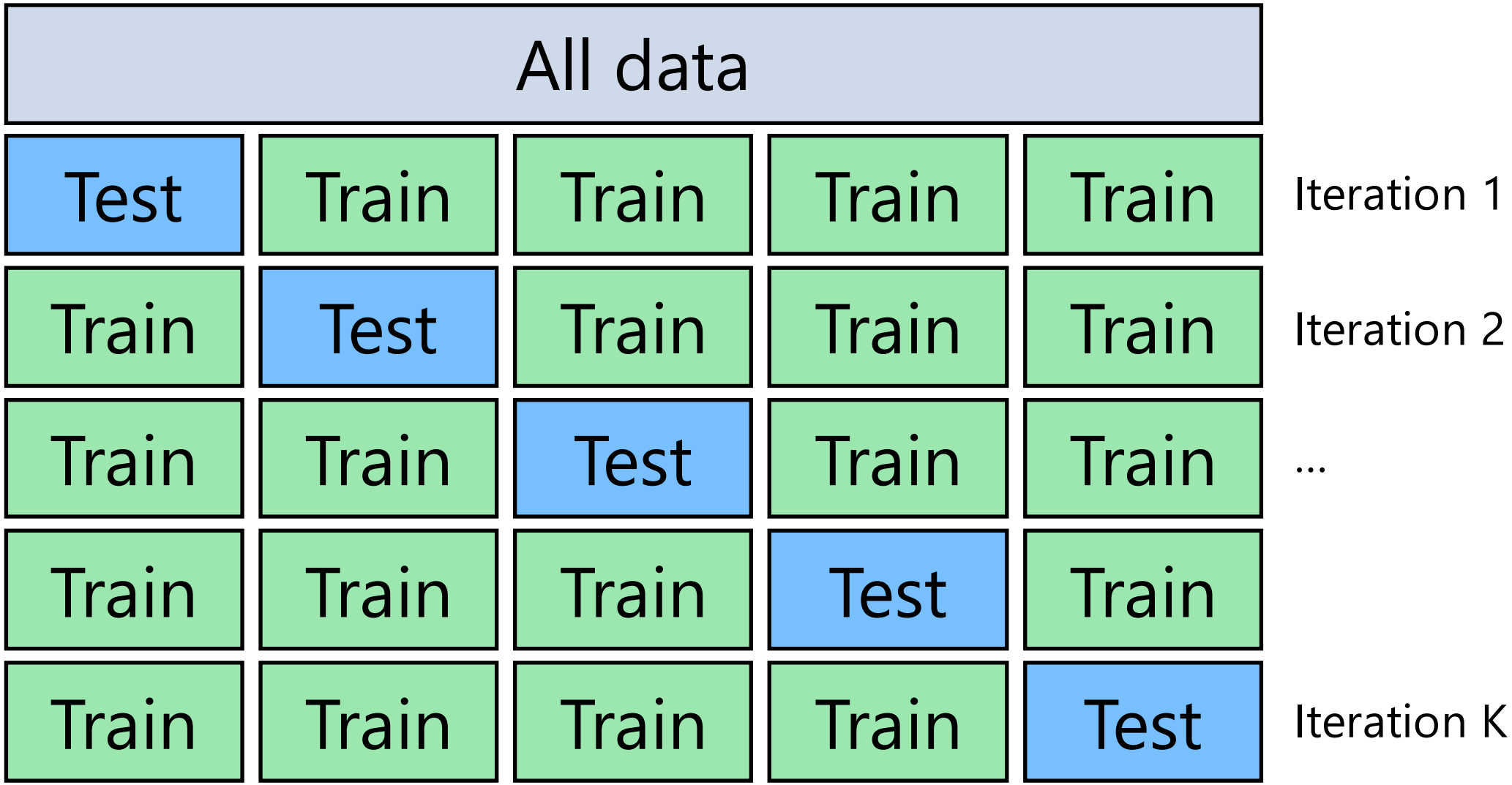
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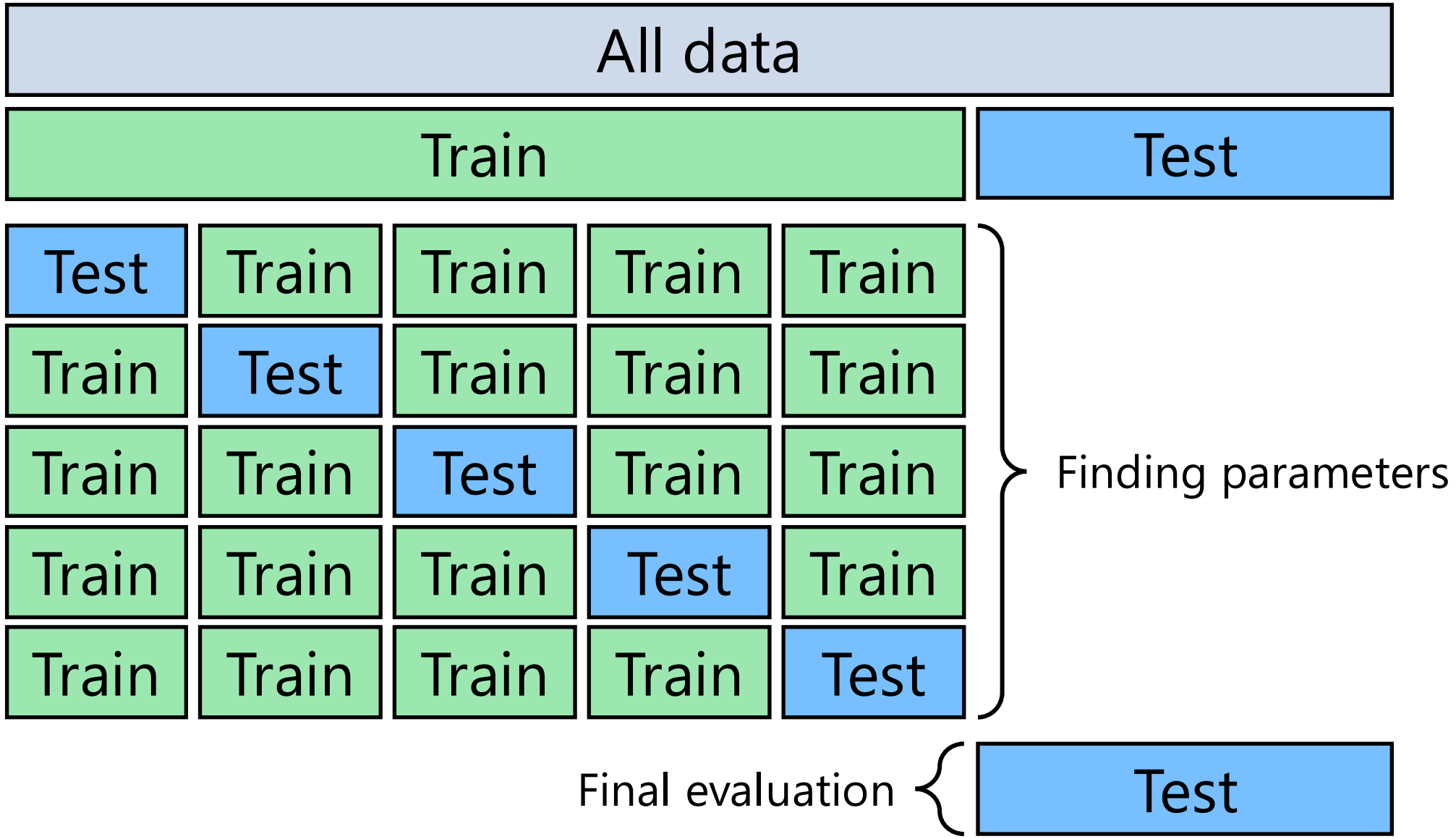
Cross-validation



K-fold cross-validation



Hyperparameter tuning



Learning check

1. Why can we not judge a model by its error on the training data?

a. the model has already seen these data, so the error looks too good

b. the training error is always zero

c. the training data are too large

d. we always can

2. We pick the best of 50 hyperparameter values by their error on the test set. This test error is then …

a. exact

b. too optimistic: we chose the value that happened to be lucky on this test set

c. too pessimistic

d. always zero

3. 5-fold cross-validation: how many times is the model trained?

a. 1

b. 5

c. 4

d. 10

4. In 5-fold cross-validation, every object is used for testing …

a. never

b. 5 times

c. 4 times

d. exactly once